Transfusion Management of Massive Haemorrhage – Obstetrics (BCUHB West)

If ongoing massive obstetric haemorrhage totalling > 1500 mls or > 40 g/l drop in [Hb]

ACTIVATE MASSIVE HAEMORRHAGE PATHWAY

Dial 3333

State "Massive Haemorrhage" and your location

Use MOH Team Co-ordinator Checklist

2 Units O Neg. PRC will be released by transfusion and brought to the stated location by a theatre porter

STOP HAEMORRHAGE

RESUSCITATE

- ABC approach
- 6L O2 via facemask
- Ensure adequate breathing
  - if hypoventilating / apnoea then assist breathing using AMBU bag
- Apply 15 degree left lateral tilt/Cardiff wedge
  - to correct aortocaval compression (for APH)
- Apply head down tilt if possible
- Assess pulse, BP & capillary refill
- Insert 2 x 14 or 16G iv cannulae & take bloods (or 2 x Intraosseous access if iv access not possible)
- Commence iv infusion of crystalloid or colloid
  - 1 litre over 5 minutes
  - recommended max. vol. of crystalloid = 2 litre
  - recommended max. vol. of colloid = 1.5 litre
- Permissive hypotension
  - aim for systolic BP of 80 mmHg until haemorrhage controlled (or max. time limit of 90 minutes)

GET SENIOR HELP

Give 2nd Tranexamic Acid 1g iv bolus

If catastrophic haemorrhage give warmed O Neg. PRC immediately

- **Take bloods :**
  - **Send to lab : XM, FBC, U+E, Ca2+ PT, APTT, fibrinogen,**
  - **NPT: HemoCue, *ABG & 2nd ROTEM***
- **Contact transfusion lab. urgently** (Ext. 5880 (Office hours) or Pager No. 043 (OOH)) and request **4 units PRC** (indicate urgency)
- **Optimise coagulation** (review ROTEM & platelet count)

FIBTEM A5
- 5 - 8mm → Give 4g Fibrinogen concentrate (or request 10 units (2 x 5 unit bags) Cryoprecipitate)
- < 5mm → Give 6g Fibrinogen concentrate (or request 15 units (3 x 5 unit bags) Cryoprecipitate)

EXTEM CT >75 sec. → Request 4 units (15ml/Kg) FFP

Platelet count < 75 → Request 1 – 2 doses Platelets

(Check early if platelets in stock . If not ,order early via blue light from WBS. If ROTEM not available, request 4 units FFP & 10 units (2 bags) Cryprecipitate

- **Consider Cell Salvage (**1 unit PRC = 250ml salvaged blood) (If Rh Neg. give 1500 unit Anti-D and repeat Kleihauer 30 min after transfusion of cell salvaged blood)
- **If ongoing haemorrhage despite normal ROTEM results and platelet count, consider requesting recombinant factor VIIa** (discuss with Consultant Haematologist)

TRANSFUSE BLOOD PRODUCTS & REASSESS PATIENT

Ongoing Haemorrhage or Aims of transfusion therapy not achieved

Haemostasis plus Aims of transfusion therapy achieved

Arrange definitive control of bleeding
Consider calling for :
- **Gynaecological Oncologist**
- **Vascular surgeon**
- **Interventional radiologist**

STAND DOWN
- **Inform lab**
- **Return unused components**
- **Complete documentation for all blood products (including audit proforma)**
- **Transfer to obs. HDU or ICU**
- **Consider thromboprophylaxis when patient stable**

Haemorrhage Control

- Follow APH and PPH pathways as appropriate.
- Copies are laminated on the labour ward haemorrhage trolley

For PPH

- Get the PPH drug box from the fridge
- Use the PPH checklist documentation

Prevent Hypothermia

- Use fluid warming device
- Used forced air warming blanket

Limitations of ROTEM

ROTEM does not :
- Detect the effect anti-platelet drugs e.g asprin & clopidogrel
- Detect the effect of the Von Willebrand factor
- Detect the effect of anticoagulants e.g. LMWH, warfarin or glycoprotein IIb/IIIa inhibitor drugs e.g Tirofiban

If these factors are present <u>do not</u> perform ROTEM

1 dose Cryoprecipitate = 10 units = 2 x 5 unit bags

Abbreviations

PRC - Packed Red Cells
FFP- Fresh Frozen plasma
ABG – Arterial Blood Gas
FBC – Full Blood Count
U+E – Urea & Electrolytes
PT - Prothrombin Time
APTT – Activated partial thromboplastin time
XM – Crossmatch
NPT – Near Patient Testing
ROTEM - Rotational Thromboelastometry
WBS – Welsh Blood Service

Monitoring

- ECG
- NIBP
- Pulse oximetry
- Urine output
- Core temperature
- CTG (for APH)

Consider :
- Arterial line
- CVP line
- Oesophageal doppler

Aims for therapy

Hb	80-100g/L
Platelets	>75 x 10^9/L
FIBTEM A5	> 9 mm
EXTEM CT	< 75 seconds
PT ratio	< 1.5
APTT ratio	<1.5
Fibrinogen	> 2g/L
Ca^{2+} (ionised) (from ABG)	>1.13mmol/L
Temp	> 36°C
pH	> 7.35 (on ABG)
Serum lactate	< 2 mmol/L
Urine output	> 0.5ml/Kg/hr

If Ca^{2+} (ionised) < 1.13mmol/l give 10 mls 10% Calcium chloride over 10 mins

If K^+ >6mmol/l give 10 mls 10% Calcium gluconate or chloride over 10 mins plus insulin with glucose

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